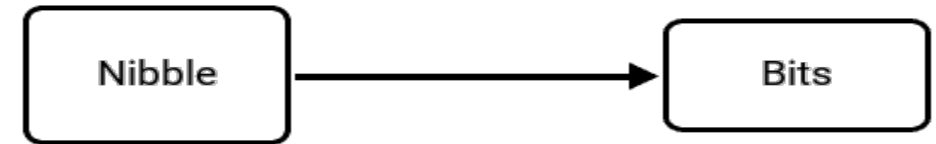




Swap Two Nibbles in a Byte

What is Nibbles



In computing, a nibble is a unit of data that consists of 4 bits. The term "nibble" is a contraction of "nybble," with the "ny" spelling emphasizing the "4" in binary (since 4 is written as "nybble" in German).

In a binary system, each digit is represented by a bit, which can be either 0 or 1. A nibble represents a group of 4 bits, allowing it to have 16 possible combinations (2^4). Each combination can represent a value from 0 to 15 in decimal or from 0000 to 1111 in binary.

In byte-oriented systems, a byte typically consists of 8 bits. Therefore, a byte can hold two nibbles. Each nibble within a byte can be manipulated or used independently.

Swapping nibbles in a byte refers to exchanging the positions of the upper nibble (the higher 4 bits) and the lower nibble (the lower 4 bits) within the byte.

Simple Concept

The concept of swapping nibbles in a byte is based on the idea of reordering the 4-bit groups within an 8-bit value. By swapping the positions of the upper nibble and the lower nibble, we effectively change the arrangement of the bits within the byte.

Here's an example to illustrate the concept:

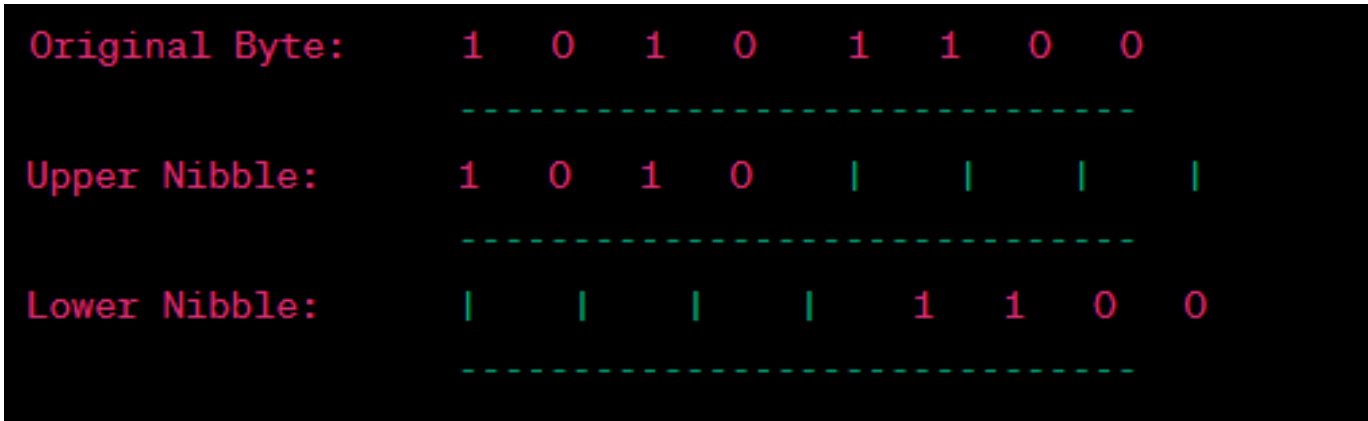
Let's consider a byte ``1010 1100`` in binary, which is represented as ``AC`` in hexadecimal notation. In this byte, the upper nibble is ``1010`` (decimal 10) and the lower nibble is ``1100`` (decimal 12).

When we swap the nibbles, the upper nibble becomes ``1100`` (decimal 12) and the lower nibble becomes ``1010`` (decimal 10). The resulting byte is ``1100 1010``, which is represented as ``CA`` in hexadecimal notation.

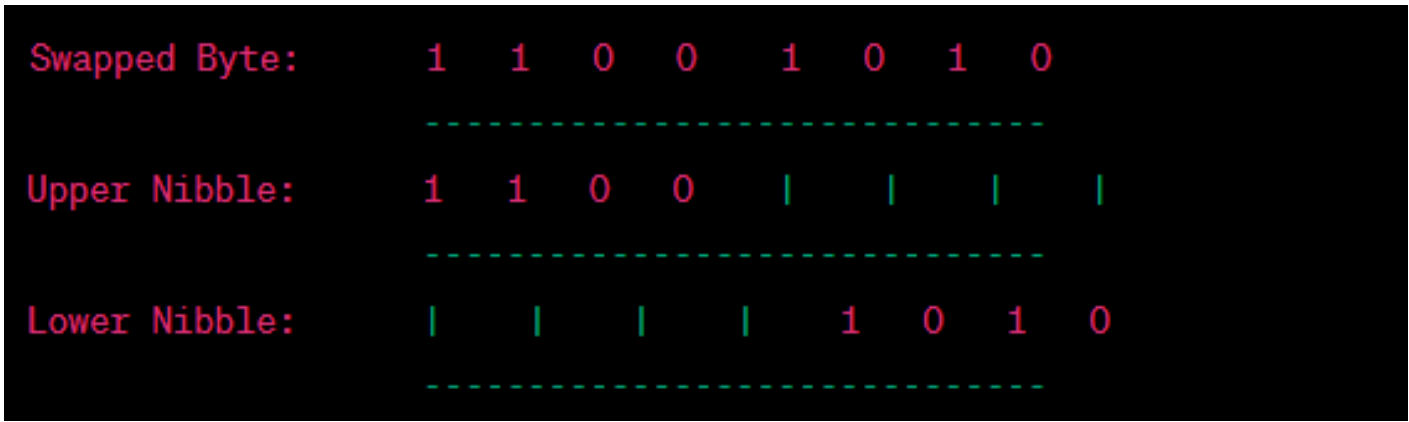
So, the concept of swapping nibbles involves rearranging the bits within a byte by exchanging the positions of the upper and lower nibbles.

Swapping nibbles can be useful in certain applications where the specific arrangement or order of the bits within a byte is important for processing or interpretation.

Consider the original byte 1010 1100:



After swapping the nibbles, the resulting byte 1100 1010 is obtained:



In this representation, each digit represents a bit within the byte. The upper nibble consists of the first four bits, and the lower nibble consists of the last four bits. By swapping the nibbles, we essentially interchange the positions of the bits within the byte.

Steps For Swap Nibbles

1. Define the function `swap_nibbles(byte)` that takes a byte as input.
2. Create a bitmask `0xF0` (11110000 in binary) to extract the upper nibble of the byte. Perform a bitwise AND operation between `byte` and the bitmask to isolate the upper nibble. Store the result in the variable `upper_nibble`.
3. Create a bitmask `0x0F` (00001111 in binary) to extract the lower nibble of the byte. Perform a bitwise AND operation between `byte` and the bitmask to isolate the lower nibble. Store the result in the variable `lower_nibble`.
4. Shift the `lower_nibble` 4 positions to the left using the left shift operator `<<`. This aligns the bits of the lower nibble to the higher positions.
5. Shift the `upper_nibble` 4 positions to the right using the right shift operator `>>`. This aligns the bits of the upper nibble to the lower positions.
6. Perform a bitwise OR operation (`|`) between the shifted `lower_nibble` and `upper_nibble` to combine them into the swapped byte. Store the result in the variable `swapped_byte`.
7. Return the value of `swapped_byte`.
8. Example usage: Assign a byte value, such as `0xAB` (1010 1011 in binary), to the variable `byte`. Call the `swap_nibbles` function with `byte` as the argument and assign the result to the variable `swapped_byte`.
9. Print the original byte and the swapped byte using the `bin()` function to display them in binary format.

```
1. public class NibbleSwap {
2.     public static byte swapNibbles(byte b) {
3.         // Extract the upper and lower nibbles
4.         byte upperNibble = (byte) ((b & 0xF0) >>> 4);
5.         byte lowerNibble = (byte) (b & 0x0F);
6.
7.         // Shift the nibbles and combine them
8.         byte swappedByte = (byte) ((lowerNibble << 4) | upperNibble);
9.
10.        return swappedByte;
11.    }

12.    public static void main(String[] args) {
13.        byte byteValue = (byte) 0xAB; // 1010 1011 in binary
14.        byte swappedByte = swapNibbles(byteValue);

15.        System.out.println("Original byte: " + Integer.toBinaryString(byteValue & 0xFF));
16.        System.out.println("Swapped byte: " + Integer.toBinaryString(swappedByte & 0xFF));
17.    }
18. }
```



THANK YOU